

Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-3580
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Proper spacing/tabs used throughout the code	Yes	Consistent Python indentation and formatted frontend code.
2	Proper File Structure	Files and folders are logically organized	Yes	Backend modules, templates, static files and tests are separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names are used for requests, prompts, responses and configuration.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as generate_text and task-specific handlers are
5	Code Comments	Inline and block comments explain logic	Partial	Core code is readable; additional comments can be added for complex AI logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Q&A, explanation, quiz, summary and learning-path logic are separated.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared Gemini generation is centralized in gemini_client.py.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Validation and API error handling are included for invalid

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	High Reusability Level
1	gemini_client.py	Python / Google GenAI	Q&A, quiz, summary and learning-path generation	High
2	qna.py	Python / FastAPI	/qa endpoint and Q&A workflow	High
3	quiz_module.py	Python / FastAPI	Quiz generation and answer-checking workflow	High
4	summary_module.py	Python / FastAPI	Text summarization workflow	High
5	learning_path.py	Python / FastAPI	Personalized learning recommendation workflow	High

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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear organization of application, AI modules, frontend assets and tests.
Readability	4	Descriptive names and straightforward request/response flows; some complex AI logic could use more comments.
Reusability	5	Task modules are separated and the shared Gemini client avoids duplicated API setup.
Documentation / Comments	4	README and module boundaries document the project; additional inline comments can improve maintainability.
Overall Score	4.5 / 5	Strong modular structure with good readability and reusable components; minor documentation improvements remain.

Implementation Summary:

Area	EduGenie Implementation
Backend	FastAPI REST API with separate task modules and validation.
Frontend	HTML, CSS and JavaScript interface using task-specific forms and result displays.
AI Integration	Google Gemini API through a shared client module; local LaMini-Flan-T5 model for concept explanation.
Testing	API and module tests cover validation and mocked AI workflows.
Maintainability	Configuration is environment-based and AI responsibilities are separated into reusable modules.